

Problem I

Problem II

Problem (Problem II.1): Let V and W be $\mathbb{K}$ -TVSs, and let $T: V \rightarrow W$ be a continuous linear map. Show that if $(x_i)_i$ is a Cauchy net in V , then $(Tx_i)_i$ is a Cauchy net in W .

Solution: Let U be a neighborhood of 0 in W . Then, the inverse image $T^{-1}(U)$ is an open neighborhood of 0 in V . There is some index k such that for all $i, j \in I$ with $i, j \geq k$, we have that $x_i - x_j \in T^{-1}(U)$. Yet, this means that $Tx_i - Tx_j \in U$ for all $i, j \in I$ with $i, j \geq k$. This gives the desired result.

Problem (Problem II.2): Let $(V, \mathcal{T}(\mathcal{N}))$ be a LCTVS, and let $(x_i)_i$ be a net in V . Let $\bar{x} \in V$ be some element. Show that the following are equivalent:

- (i) the net $(x_i)_i$ converges to $\bar{x}$;
- (ii) for all $\rho \in \mathcal{N}$ and for all $\varepsilon > 0$, there exists $k \in I$ such that for all $i \in I$ with $i \geq k$, one has $\rho(x_i - \bar{x}) \leq \varepsilon$;
- (iii) for all continuous seminorms ρ and for all $\varepsilon > 0$, there exists $k \in I$ such that for all $i \in I$ with $i \geq k$, one has $\rho(x_i - \bar{x}) \leq \varepsilon$.

Solution: Since translation by a vector is a homeomorphism, we will assume that $\bar{x} = 0$ and prove the results accordingly.

Suppose $(x_i)_i \rightarrow 0$. If ρ is a defining seminorm, then we observe that the subbasic open set

$$U(0, \rho, \varepsilon) = \{y \in V : \rho(y) < \varepsilon\}$$

is an open neighborhood of 0. In particular, this means there is some $k \in I$ such that for all $i \in I$ with $i \geq k$, one has $x_i \in U(0, \rho, \varepsilon)$, or that $\rho(x_i) < \varepsilon$. This gives (ii).

To show (iii) from (ii), one observes that if ρ is a continuous seminorm, then there are defining seminorms $\rho_1, \dots, \rho_n$ and a constant C such that

$$\rho(x) \leq C \max\{\rho_1(x), \dots, \rho_n(x)\}.$$

Given $\varepsilon > 0$, for each $j = 1, \dots, n$ we select $k_j \in I$ such that for all $i_j \in I$ satisfying $i_j \geq k_j$, one has $\rho(x_{i_j}) < \varepsilon/C$. There is some k such that $k \geq k_j$ for each $j = 1, \dots, n$. Then, for any $i \in I$ with $i \geq k$, one has $\rho(x_i) \leq \varepsilon$, yielding (iii).

Finally, assuming (iii) holds, we observe that for any neighborhood U of 0, there is some continuous seminorm ρ and some $\varepsilon > 0$ such that

$$\begin{aligned} U(0, \rho, \varepsilon) &= \{y \in V : \rho(y) < \varepsilon\} \\ &\subseteq U. \end{aligned}$$

In particular, this means that there exists some $k \in I$ such that for all $i \in I$ with $i \geq k$, one has $\rho(x_i) \leq \varepsilon/2 < \varepsilon$ by the assumption of (iii). In particular, this means for all $i \geq k$, one has $x_i \in U(0, \rho, \varepsilon) \subseteq U$. Thus, the net $(x_i)_i$ converges to x .

Problem (Problem II.3): Let $(V, \mathcal{T}(\mathcal{N}))$ be a locally convex topological vector space, and let $(x_i)_i$ be a net in V . Show that the following are equivalent:

- (i) the net $(x_i)_i$ is Cauchy;
- (ii) for all $\rho \in \mathcal{N}$ and for all $\varepsilon > 0$, there exists $k \in I$ such that for all $i, j \in I$ with $i, j \geq k$, one has $\rho(x_i - x_j) \leq \varepsilon$;
- (iii) for all continuous seminorms ρ and for all $\varepsilon > 0$, there exists $k \in I$ such that for all $i, j \in I$ with $i, j \geq k$, one has $\rho(x_i - x_j) \leq \varepsilon$.

k , one has $\rho(x_i - x_j) \leq \varepsilon$.

Solution: Suppose $(x_i)_i$ is Cauchy. Let ρ be a defining seminorm for the topology on V . Then, the subbasic open set

$$U(0, \rho, \varepsilon) = \{y \in V : \rho(y) < \varepsilon\}$$

is an open neighborhood of 0. Therefore, there is k such that for all $i, j \geq k$, one has $x_i - x_j \in U(0, \rho, \varepsilon)$. Equivalently, we have that $\rho(x_i - x_j) \leq \varepsilon$, giving (ii).

Now, suppose (ii) holds. Then, if ρ is a continuous seminorm, there are defining $\rho_1, \dots, \rho_n$ and there is a constant C such that for all $x \in V$, one has

$$\rho(x) \leq C \max\{\rho_j(x) : j = 1, \dots, n\}.$$

Since (ii) holds, it follows that there is some k such that for all $i, j \geq k$, one has $\rho_1(x_i - x_j), \dots, \rho_n(x_i - x_j) \leq \varepsilon/C$, whence $\rho(x_i - x_j) \leq \varepsilon$.

Finally, suppose (iii) holds. Let U be an open neighborhood of 0. Then, there is some continuous seminorm ρ and some $\varepsilon > 0$ such that $U(0, \rho, \varepsilon) \subseteq U$. It follows that there exists $k \in I$ such that $\rho(x_i - x_j) < \varepsilon$ for all $i, j \geq k$. It follows that $x_i - x_j \in U(0, \rho, \varepsilon) \subseteq U$, so that $(x_i)_i$ is Cauchy.

Problem (Problem II.4): Let V be a Banach space. Show that this space satisfies the new definition of complete.

Solution: Suppose $(x_i)_i$ is a Cauchy net. We will assume without loss of generality that the directed set I is infinite. Notice that Banach spaces are first countable. That is, for any neighborhood U of 0, there is some natural number $n \geq 1$ such that $U(0, 1/n) \subseteq U$.

Using the countable family of neighborhoods $\{U(0, 1/n) : n \geq 1\}$, we may extract a sequence in the net as follows. Select k_1 such that for all $i, j \geq k_1$, we have $\|x_i - x_j\| \leq 1$; thus, for all $i \geq k_1$, we have $\|x_i - x_{k_1}\| \leq 1$. Since $I \setminus \{k_1\}$ is still infinite, it follows that we may find k'_2 such that for all $i \geq k_2$, $\|x_i - x_{k_2}\| \leq \frac{1}{2}$, so if we let k_2 be an upper bound for $\{k_1, k'_2\}$, then for all $i \geq k_2$, we have that $\|x_i - x_{k_2}\| \leq 1/2$. Inductively, given $\{k_1, \dots, k_n\}$, we may find $k_{n+1} \in I$ since I is infinite such that for all $i \geq k_{n+1}$, we have $\|x_i - x_{k_{n+1}}\| \leq \frac{1}{n+1}$. This gives us a countable subnet $(x_{k_j})_{j \in \mathbb{N}}$ that satisfies

$$\|x_{k_{n+1}} - x_{k_n}\| \leq \frac{1}{n},$$

which yields that this countable subnet (i.e., sequence) is Cauchy. Since any Cauchy net contains a countable Cauchy subnet (i.e., a Cauchy sequence), it follows that any Cauchy net converges.

Problem (Problem II.5): Show that a TVS is a Fréchet space if and only if it is locally convex, metrizable, and complete.

Solution: We will show that if V is locally convex and metrizable, with countable family of defining seminorms $\{\rho_k : k \in \mathbb{N}\}$ and distance metric given by

$$d(x, y) = \sum_{j=0}^{\infty} 2^{-j} \min(1, \rho_j(x - y))$$

that induces the topology, then a net $(x_i)_i$ is d -Cauchy if and only if it is Cauchy as a net.

Suppose $(x_i)_i$ is d -Cauchy. Let ρ_j be a defining seminorm. Observe that

$$\min(1, \rho_j(x - y)) \leq 2^j d(x - y).$$

Let $\varepsilon > 0$. Then, since $(x_i)_i$ is d -Cauchy, there is some k such that $2^j d(x_i - x_j) < \min(1, \varepsilon)$ for all $u, v \geq k$. Therefore, since

$$\min(1, \rho_j(x_u - x_v)) \leq 2^j d(x_u - x_v) < \min(1, \varepsilon),$$

it follows that $\rho_j(x_u - x_v) < \min(1, \varepsilon) \leq \varepsilon$ for all $u, v \geq k$, so that $(x_i)_i$ is Cauchy as a net.

Now, suppose $(x_i)_i$ is Cauchy as a net. Let $\varepsilon > 0$, and let K be sufficiently large such that $\sum_{j=K}^{\infty} 2^{-j} < \varepsilon/2$. Notice that

$$\rho = \sum_{j=0}^{K-1} 2^{-j} \rho_j$$

is a continuous seminorm. Therefore, there is some k_0 such that for all $u, v \geq k_0$, we have $\rho(x_u - x_v) < \varepsilon/2$. This gives, for all $u, v \geq k_0$,

$$\begin{aligned} d(x_u, x_v) &= \sum_{j=0}^{K-1} 2^{-j} \min(1, \rho_j(x_u - x_v)) + \sum_{j=K}^{\infty} 2^{-j} \min(1, \rho_j(x_u - x_v)) \\ &\leq \sum_{j=0}^{K-1} 2^{-j} \rho_j(x_u - x_v) + \sum_{j=K}^{\infty} 2^{-j} \\ &< \varepsilon. \end{aligned}$$

Therefore, $(x_i)_i$ is d -Cauchy.

In particular, this gives that a space is Fréchet if and only if d -Cauchy nets converge if and only if Cauchy nets converge if and only if the space is complete as a topological vector space.

Problem (Problem II.6): For any $\omega \in [0, \infty)^{\mathbb{N}}$ and $x = (x_n)_n \in s_0(\mathbb{N}, \mathbb{K})$, we let $\|x\|_{\omega} = \sup_{n \geq 0} \omega_n |x_n|$. Show that these seminorms define the topology of $s_0(\mathbb{N}, \mathbb{K})$.

Solution: Let $\mathcal{T}_1$ be the original topology generated by all the seminorms of $s_0(\mathbb{N}, \mathbb{K})$, and let $\mathcal{T}_2$ be this new topology. If $(x_i)_i$ converges to 0 in $\mathcal{T}_1$, then since the seminorms $\|x\|_{\omega}$ as defined in this question are seminorms on $s_0(\mathbb{N}, \mathbb{K})$, it follows that they must converge in $\mathcal{T}_2$.

In the reverse direction, notice that for any $\rho \in \text{SN}(s_0(\mathbb{N}, \mathbb{K}))$, we consider the sequence(s) given by $\omega^{(n)} = \rho(e_n)e_n$. Then, it follows that for all $x \in X$, we have

$$\begin{aligned} \rho(x) &\leq \sum_{n=0}^{\infty} \rho(e_n) |x_n| \\ &= \sum_{n=0}^{\infty} \left(\sup_{j \in \mathbb{N}} \omega_j^{(n)} |x_n| \right). \end{aligned}$$

The latter gives the desired upper bound that establishes the continuity of any such ρ .

Problem (Problem II.7): Let $y = (y_n)_n \in \mathbb{K}^{\mathbb{N}}$ such that for all $(\lambda_n)_n \in (0, \infty)^{\mathbb{N}}$, there exists $N \geq 0$ such that for all $n \geq N$, $|y_n| \leq \lambda_n$. Show that y is finitely supported.

Solution: Suppose y were not finitely supported. Then, there is a subsequence $(y_{n_k})_k$ such that $(|y_{n_k}|)_k \in (0, \infty)^{\mathbb{N}}$. Consider now the sequence $(\lambda_n)_n$ defined by

$$\lambda_n = \begin{cases} \frac{1}{n+1} & n \neq n_k \text{ for any } k \\ \frac{1}{2} |y_{n_k}| & n = n_k. \end{cases}$$

For any $N \in \mathbb{N}$, there exists some $n_k \geq N$ since $(n_k)_k$ is an unbounded sequence, so that $\lambda_{n_k} < |y_{n_k}|$. Thus, any such infinitely supported sequence y cannot satisfy the hypothesis.

Problem (Problem II.8): Let $(x_i)_i$ be a net in $s_0(\mathbb{N}, \mathbb{K})$. We will write $x_i = (x_i(n))_{n \geq 0}$ for each $i \in I$. Suppose $(x_i)_i$ is Cauchy in $s_0(\mathbb{N}, \mathbb{K})$. Show that there is a sequence $y = (y_n)_n \in \mathbb{K}^{\mathbb{N}}$ such that for all $n \geq 0$, the net $(x_i(n))_i$ converges to y_n in $\mathbb{K}$.

Solution: The seminorms $\rho_n(x_i) = |x_i(n)|$ are continuous in $s_0(\mathbb{N}, \mathbb{K})$. It follows that for each n , we have that the nets $(x_i(n))_{i \in I}$ are Cauchy in $\mathbb{K}$, so that they converge to $y(n)$. That is, we may define $(y(n))_n$ by taking pointwise limits.

We claim that $(x_i)_i \rightarrow y$ in $s_0(\mathbb{N}, \mathbb{K})$. Let $\omega \in [0, \infty)^{\mathbb{N}}$. For each n , we have some k such that for all $j \geq k$, we have $|x_j(n) - y(n)| < \varepsilon/2\omega(n)$. It follows that for each n , we have $\sup_{j \geq k} \omega(n)|x_j(n) - y(n)| < \varepsilon/2$, whence

$$\sup_{n \in \mathbb{N}} \sup_{j \geq k} \omega(n)|x_j(n) - y(n)| \leq \varepsilon/2,$$

so that for all $j \geq k$, we have that $\sup_{n \in \mathbb{N}} \omega(n)|x_j(n) - y(n)| < \varepsilon$. Thus, $(x_i)_i \rightarrow y$ in $s_0(\mathbb{N}, \mathbb{K})$.

Problem (Problem II.9): With the same hypotheses as in the previous question, show that $y \in s_0(\mathbb{N}, \mathbb{K})$ and $(x_i)_i \rightarrow y$ in the topology of $s_0(\mathbb{N}, \mathbb{K})$.

Solution: Rewriting the Cauchy hypothesis, we have that for all $\omega \in [0, \infty)^{\mathbb{N}}$ and for all $\varepsilon > 0$, there exists some $k \in I$ such that for all $i, j \geq k$ and for all $n \geq 0$, we have that $|x_i(n) - x_j(n)| \leq \frac{\varepsilon}{\omega_n}$. (For the sake of argument, we will always assume that $\omega_n > 0$.) Having defined $y(n)$ as the pointwise limit of $(x_i(n))_i$, we may take the limit with respect to j , yielding that for all $n \geq 0$ and for all $i \geq k$, we have

$$|x_i(n) - y(n)| \leq \frac{\varepsilon}{\omega_n}.$$

Now, observe that for all $i \geq k$, we have

$$\begin{aligned} |y(n)| &\leq |x_i(n) - y(n)| + |x_i(n)| \\ &\leq \frac{\varepsilon}{\omega_n} + |x_i(n)|. \end{aligned}$$

Let N be the first index for which $x_i(n) = 0$ for all $n \geq N$. Then, we have that $|y(n)| \leq \frac{\varepsilon}{\omega_n}$. Letting $\lambda_n = \frac{\varepsilon}{\omega_n}$, it follows that, since ω_n is any (nonzero) sequence, it follows that $(y(n))_n$ satisfies the hypotheses of II.7, so that $y \in s_0(\mathbb{N}, \mathbb{K})$. Since, for all $\omega \in [0, \infty)^{\mathbb{N}}$ and for all $n \geq 0$, we have that $\omega_n|x_i(n) - y(n)| \leq \varepsilon$ whenever $i \geq k$, it follows that $(x_i)_i \rightarrow y$ in $s_0(\mathbb{N}, \mathbb{K})$.

Problem III

Problem (Problem III.1): We say a neighborhood U of the origin in a topological vector space $(V, \mathcal{T})$ is balanced if and only if for all $x \in U$ and all $\lambda \in \mathbb{K}$ with $|\lambda| \leq 1$, we have $\lambda x \in U$. Show that for all open neighborhoods $U \in \mathcal{O}_V(0)$, there exists $B \in \mathcal{O}_V(0)$ with $B \subseteq U$ and B balanced.

Solution: Let U be an open neighborhood of 0. Then, since the scalar multiplication map $m: \mathbb{K} \times V \rightarrow V$ is continuous, it follows that $m^{-1}(U)$ is an open subset of $\mathbb{K} \times V$. Therefore, there is a basic open subset $U(0, \varepsilon) \times W \subseteq m^{-1}(U)$, where

$$U(0, \varepsilon) = \{z \in \mathbb{K} : |z| < \varepsilon\}.$$

It follows that

$$m(U(0, \varepsilon) \times W) = \bigcup_{|z| < \varepsilon} zW$$

is a balanced open subset of U containing 0. This yields our desired result.

Problem (Problem III.2): Show that, in a topological vector space, if two subsets A and B are bounded, then $A \cup B$ is bounded.

Solution: Let $U \in \mathcal{O}_V(0)$ be a neighborhood of 0. Then, there is a balanced subset $W \in \mathcal{O}_V(0)$ with $W + W \subseteq U$, where $W + W$ denotes the Minkowski sum of W with itself. Such a neighborhood can be found by obtaining a balanced subset W' of U , then taking $W = \frac{1}{2}W'$.

There is ε_1 such that $\varepsilon_1 A \subseteq W$ and $\varepsilon_2 B \subseteq W$. Taking $\varepsilon = \min(\varepsilon_1, \varepsilon_2)$, we observe that $\frac{\varepsilon}{\varepsilon_1} \leq 1$ and $\frac{\varepsilon}{\varepsilon_2} \leq 1$. Therefore, $\varepsilon A \subseteq \frac{\varepsilon}{\varepsilon_1} W \subseteq W$ since W is balanced. Similarly, $\varepsilon B \subseteq \frac{\varepsilon}{\varepsilon_2} W \subseteq W$. Therefore, $\varepsilon(A \cup B) = \varepsilon A \cup \varepsilon B \subseteq W \cup W = W$.

Problem (Problem III.3): We define sets of nonnegative sequences with specified behavior at infinity as follows

$$s_+(\mathbb{N}, \mathbb{K}) = s(\mathbb{N}, \mathbb{K}) \cap [0, \infty)^\mathbb{N}$$

$$s_{0,+}(\mathbb{N}, \mathbb{K}) = s_0(\mathbb{N}, \mathbb{K}) \cap [0, \infty)^\mathbb{N}.$$

Show that $A \subseteq s(\mathbb{N}, \mathbb{K})$ is bounded if and only if there exists $\omega \in s_+(\mathbb{N}, \mathbb{K})$ such that for all $x = (x_n)_n \in A$ and for all $n \geq 0$, we have $|x_n| \leq \omega_n$. Show that a similar result holds for the finitely supported sequences.

Solution: Let A be a bounded subset of $s(\mathbb{N}, \mathbb{K})$. Since $s(\mathbb{N}, \mathbb{K})$ with the given family of seminorms is metrizable with the metric

$$d(x, y) = \sum_{k=0}^{\infty} 2^{-k} \min(1, \rho_k(x - y)),$$

there is some $r > 0$ such that $A \subseteq U(0, r)$. The latter set consists of all sequences $(u_n)_n$ such that

$$d(0, u) = \sum_{k=0}^{\infty} 2^{-k} \max(1, \rho_k(u)) < r.$$

Let $r_0 = \inf_r \{r : A \subseteq U(0, r)\}$. Then, we choose $\omega_n \geq 0$.